Questions 33 to 37

(I) T stays as PG; S stays in hostel.
U stays in hostel and Q stays at home.
Now, R does not stay as PG. So, P stays as PG.
Clearly, R stays at home.

(II) S studies Physics; R studies Philosophy and T studies Mathematics. Now, P who stays as PG does not study Statistics or History. So, P studies English.

	Place of stay	Subject
P	PG	English
Q	Home	Statistics or History
R	Home	Philosophy
S	Hostel	Physics
T	PG	Mathematics
U	Hostel	Statistics or History

33. (*e*) : P studies English.
34. (*a*) : Clearly the incorrect combination is English — Hostel.
35. (*b*) : S stays in hostel and R stays at home.
36. (*c*) : Q studies History or Statistics.
37. (*b*) : Q and R stay at home.

Questions 38 to 42

(I) Kunal stays close to school. So, the other three — Rohit, Ashish and John stay far from the school.

(II) Rohit, Kunal and Ashish do not study in class VI. So, John studies in class VI.
Rohit and Kunal are not good at all subjects and John is in class VI. So, Ashish is good at all subjects and studies in class V.
Clearly, Rohit and Kunal study in class IV.

(III) Rohit is good at Mathematics, Kunal is weak in Mathematics. Ashish is good at all the subjects. Clearly, John is weak in all the subjects.

	Stay	Class	Good at	Weak in
Rohit	Far	IV	Mathematics	Hindi, Science, Social Science
Kunal	Close	IV	Hindi, Science, Social Science	Mathematics
Ashish	Far	V	All subjects	
John	Far	VI		All subjects

38. (*c*) : Ashish is good at all the subjects.
39. (*d*) : John is weak in all the subjects.
40. (*b*) : Kunal and Ashish are good at Hindi.
41. (*a*) : Rohit and Ashish are good at Mathematics.
42. (*d*) : Other than Rohit and Ashish, John stays far from school.

Questions 43 to 45

(I) D plays Polo; E plays Badminton; A plays Cricket. C does not play Tennis or Hockey. So, C plays Volleyball.
B does not play Tennis. So, he plays Hockey. Clearly, F plays Tennis.

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Questions 24 to 28

The given information can be analysed as follows :

Let 'O' denote optional and 'C' denote compulsory.

	A	B	C	D	E	F
History	C	C	C	O	–	–
Physics	–	–	–	–	C	O
Mathematics	–	–	O	–	O	C
English	O	–	–	C	–	–
Chemistry	–	O	–	–	–	–

One compulsory subject of F is Mathématics. F has Physics as one of the subjects. So, Physics is optional of F. F has Mathematics as optional and Physics as one subject.

So, Physics is the compulsory subject of E. A and D have the same subjects — History and English. D has History as optional subject and so English is the compulsory subject of D. Subjects of A and D are reverse in regard of optional and compulsory.

So, A has History as compulsory subject and English as optional.

Chemistry is the optional subject of only one teacher. So, it is the optional of B, which only remains.

We know that History is the compulsory subject of three teachers. So, it is compulsory for A, B and C.

D is the teacher having English as her compulsory subject.

So, D is the only female teacher.

24. (*a*) : C's compulsory subject is History.

25. (*d*) : D is the only female teacher.

26. (*e*) : E has same subjects as those of F but the compulsory and optional subjects of E are reverse of those of F. So, the answer is 'none of these'.

27. (*c*) : Clearly, E has the same subject combination as that of F.

28. (*d*) : A, B and C have History as the compulsory subject.

Questions 29 to 32

(I) **Choice :** Jayant likes Chocolates; Tanmay likes Bengali sweets. Namita does not like Dry fruits or Ice cream. So, Namita likes Pastries. Asha does not like Ice cream. So, she is fond of Dry fruits. Finally, Kamal likes Ice cream.

(II) **Date of Birth :** The one who likes Pastries *i.e.*, Namita is born in the middle of months given *i.e.*, in March. Tanmay is born in next month after Namita *i.e.*, in April. Jayant's birthday is in February.

	Choice	Date of birth
Jayant	Chocolates	February
Kamal	Ice cream	January or May
Namita	Pastries	March
Asha	Dry fruits	January or May
Tanmay	Bengali sweets	April

29. (*d*) : The choice of Asha is Dry fruits.

30. (*e*) : The correct combination for Jayant is February — Chocolates.

31. (*a*) : The choice of Kamal is Ice cream.

32. (*c*) : Kamal was born in January or May.

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(II) D has yellow car; E has white car; B has a red car; A has a black car.
Now, C does not have a blue car. So, colour of C's car is green.
Clearly, F owns a blue car.

	Game	Colour of Car
A	Cricket	Black
B	Hockey	Red
C	Volleyball	Green
D	Polo	Yellow
E	Badminton	White
F	Tennis	Blue

43. (*b*) : C plays Volleyball.

44. (*b*) : F owns a blue car.

45. (*b*) : Clearly, the incorrect combination is Green — Tennis.

46. (*c*) : The four vocal musicians and one actress in the group are given. Two dancers are Shailja and Tanuja. Two violinists are Girija and Vanaja. Since Jalaja and Shailja cannot be violinists, so, remaining two violinists are Tanuja and Pooja. Clearly, Tanuja is both a violinist and a dancer.

TYPE 2 : SEATING/PLACING ARRANGEMENTS

In this type of questions, some clues regarding seating or placing sequence (linear or circular) of some persons or items is given. The candidate is required to form the proper sequence using these clues and answer the questions accordingly.

Ex. 1. ***Read the following information carefully and answer the questions given below :***

Six persons A, B, C, D, E and F are sitting in two rows, three in each.
E is not at the end of any row.
D is second to the left of F.
C, the neighbour of E, is sitting diagonally opposite to D.
B is the neighbour of F.

1. Which of the following are sitting diagonally opposite to each other ?
(*a*) F and C (*b*) D and A (*c*) A and C (*d*) A and F (*e*) A and B

2. Who is facing B ?
(*a*) A (*b*) C (*c*) D (*d*) E (*e*) F

3. Which of the following are in the same row ?
(*a*) A and E (*b*) E and D (*c*) C and B (*d*) A and B (*e*) C and E

4. Which of the following are in one of the two rows ?
(*a*) FBC (*b*) CEB (*c*) DBF (*d*) AEF (*e*) ABF

5. After interchanging seat with E, who will be the neighbours of D in the new position ?
(*a*) C and A (*b*) F and B (*c*) Only B (*d*) Only A (*e*) Only C

Solution : The given information can be analysed as follows :
E is not at end. So, E must be in the middle of one of the rows.
D is second to the left of F. So, order of the row must be D — F.

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C is neighbour of E and is sitting diagonally opposite to D means C is under F in the other row *i.e.*, D — F

— E C

B is the neighbour of F.

So, the arrangement must be D B F

A E C

1. Other than D and C (given), A and F are sitting diagonally opposite to each other, as seen in the arrangement. So, the answer is (*d*).
2. Clearly, E is opposite to B in the other row. So, E is facing B and the answer is (*d*).
3. Clearly, from amongst the given alternatives, A and E are in the same row. So, the answer is (*a*).
4. Clearly, from amongst the given alternatives, D, B and F are in the same row. So, the answer is (*c*).
5. Clearly, neighbours of E are A and C. So, on interchanging the seat with E, the new neighbours of D will be A and C. So, the answer is (*a*).

Ex. 2. Eight books are kept one over the other. Counting from the top, the second, fifth and sixth books are on Plays. Two books on Plays are between two books on Composition. One book of Plays is between two books on Poetry while the book at the top of the book of Literature is a book of Composition. Which book is fourth from the top ?

(*a*) Plays (*b*) Poetry (*c*) Composition (*d*) Literature

Solution : We analyse the given information as follows :

Let C denote 'Composition', P denote 'Plays', Po denote 'Poetry' and L denote 'Literature'.

1	2	3	4	5	6	7	8
–	P	–	–	P	P	–	–
–	–	–	C	P	P	C	–
Po	P	Po	–	–	–	–	–
–	–	–	–	–	–	C	L

So, the arrangement becomes :

1	2	3	4	5	6	7	8
Po	P	Po	C	P	P	C	L

Clearly, the fourth book from the top is on Composition. So, the answer is (*c*).

Ex. 3. ***Read the following information and answer the questions that follow :***

(*i*) Six friends A, B, C, D, E and F are sitting in a closed circle facing the centre.

(*ii*) E is to the left of D.

(*iii*) C is between A and B.

(*iv*) F is between E and A.

1. Who is to the left of B ?

(*a*) A (*b*) C (*c*) D (*d*) E (*e*) None of these

2. Who is to the right of C ?

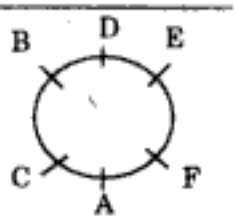
(*a*) A (*b*) B (*c*) D (*d*) E (*e*) F

3. Which of the above given statements is superfluous ?

(*a*) (*i*) (*b*) (*ii*) (*c*) (*iii*) (*d*) (*iv*) (*e*) None of these

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Solution : Clearly, in the circle the arrangement is as shown :



1. (*c*) : Clearly, D is to the left of B.

2. (*a*) : Clearly, A is to the right of C.

3. (*e*) : Since all the statements are necessary to determine the arrangement, none of them is superfluous.

EXERCISE 6B

1. Four girls are sitting on a bench to be photographed. Shikha is to the left of Reena. Manju is to the right of Reena. Rita is between Reena and Manju. Who would be second from the left in the photograph ? **(Bank P.O. 1994)**

(*a*) Reena (*b*) Shikha (*c*) Manju (*d*) Rita

2. There are five different houses, A to E, in a row. A is to the right of B and E is to the left of C and right of A. B is to the right of D. Which of the houses is in the middle ? **(C.B.I. 1995)**

(*a*) A (*b*) B (*c*) D (*d*) E

3. In a March Past, seven persons are standing in a row. Q is standing left to R but right to P. O is standing right to N and left to P. Similarly, S is standing right to R and left to T. Find out who is standing in the middle.

(*a*) P (*b*) Q (*c*) R (*d*) O

(Assistant Grade, 1996)

4. Five children are sitting in a row. S is sitting next to P but not T. K is sitting next to R who is sitting on the extreme left and T is not sitting next to K. Who are sitting adjacent to S ?

(*a*) K and P (*b*) R and P (*c*) Only P

(*d*) P and T (*e*) Insufficient information

5. Five girls are sitting in a row. Rashi is not adjacent to Sulekha or Abha. Anuradha is not adjacent to Sulekha. Rashi is adjacent to Monika. Monika is at the middle in the row. Then, Anuradha is adjacent to whom out of the following ?

(*a*) Rashi (*b*) Sulekha (*c*) Abha

(*d*) Monika (*e*) Cannot be determined

Directions (*Questions 6 to 8*) : *Read the following information carefully and answer the questions given below it :* **(Bank P.O. 1994)**

(A) There are five friends.

(B) They are standing in a row facing South.

(C) Jayesh is to the immediate right of Alok.

(D) Pramod is between Bhagat and Subodh.

(E) Subodh is between Jayesh and Pramod.

6. Who is at the extreme left end ?

(*a*) Alok (*b*) Bhagat (*c*) Subodh

(*d*) Data inadequate (*e*) None of these

7. Who is in the middle ?

(*a*) Bhagat (*b*) Jayesh (*c*) Pramod

(*d*) Subodh (*e*) Alok

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8. To find answers to the above two questions, which of the given statements can be dispensed with ?

(*a*) None (*b*) A only (*c*) B only (*d*) C only **(*e*) D only**

9. Five persons A, B, C, D and E are sitting in a row facing you such that D is on the left of C and B is on the right of E. A is on the right of C and B is on the left of D. If E occupies a corner position, then who is sitting in the centre ?

(*a*) A (*b*) B (*c*) C (*d*) D (C.B.I. 1995)

Directions (*Questions 10 to 14*) : *Study the given information carefully and answer the questions that follow :* (L.I.C. 1994)

(*i*) A, B, C, D, E, F and G are sitting on a wall and all of them are facing east.

(*ii*) C is on the immediate right of D.

(*iii*) B is at an extreme end and has E as his neighbour.

(*iv*) G is between E and F.

(*v*) D is sitting third from the south end.

10. Who is sitting to the right of E ?

(*a*) A (*b*) C (*c*) D (*d*) F (*e*) None of these

11. Which of the following pairs of people are sitting at the extreme ends ?

(*a*) AB (*b*) AE (*c*) CB (*d*) FB (*e*) Cannot be determined

12. Name the person who should change places with C such that he gets the third place from the north end.

(*a*) E (*b*) F (*c*) G (*d*) D

13. Immediately between which of the following pairs of people is D sitting ?

(*a*) AC (*b*) AF (*c*) CE (*d*) CF (*e*) None of these

14. Which of the conditions (*i*) to (*v*) given above is not required to find out the place in which A is sitting ?

(*a*) (*i*) (*b*) (*ii*) (*c*) (*iii*) (*d*) All are required (*e*) None of these

15. In the Olympic Games, the flags of six nations were flown on the masts in the following way : (Assistant Grade, 1996)

The flag of America was to the left of Indian tricolour and to the right of the flag of France. The flag of Australia was on the right of the Indian flag but was to the left of the flag of Japan, which was to the left of the flag of China. Find the two flags which are in the centre.

(*a*) India and Australia (*b*) America and India

(*c*) Japan and Australia (*d*) America and Australia

16. Mr. A, Miss B, Mr. C and Miss D are sitting around a table and discussing their trades.

(1) Mr. A sits opposite to cook.

(2) Miss B sits right to the barber.

(3) The washerman is on the left of the tailor.

(4) Miss D sits opposite Mr. C.

What are the trades of A and B ?

(*a*) Tailor and Barber (*b*) Tailor and Cook

(*c*) Barber and Cook (*d*) Washerman and Cook

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17. Sitting in a row in front of a camera, Mr. X is on the left of the person sitting in the centre but is on the right of Mr. Y. Mr. P is on the right of Mr. Z and Mr. R is on the right of Mr. P. Mr. R is the second person from the person sitting in the centre. Who is the person sitting in the centre ? **(C.B.I. 1993)**

(*a*) Mr. X (*b*) Mr. Y (*c*) Mr. Z (*d*) Mr. R

Directions (*Questions 18 to 22*) : *Study the given information carefully and answer the questions that follow :* **(U.T.I. 1993)**

(*i*) Eleven students, A, B, C, D, E, F, G, H, I, J and K are sitting in the first row of the class facing the teacher.

(*ii*) D who is to the immediate left of F is second to the right of C.

(*iii*) A is second to the right of E, who is at one of the ends.

(*iv*) J is the immediate neighbour of A and B and third to the left of G.

(*v*) H is to the immediate left of D and third to the right of I.

18. Who is sitting in the middle of the row ?

(*a*) B (*b*) C (*c*) G (*d*) I (*e*) None of these

19. Which of the following groups of friends is sitting to the right of G ?

(*a*) CHDE (*b*) CHDF (*c*) IBJA (*d*) ICHDF (*e*) None of these

20. Which of the following statements is true in the context of the above sitting arrangements ?

(*a*) There are three students sitting between D and G.

(*b*) K is between A and J.

(*c*) B is sitting between J and I.

(*d*) G and C are neighbours sitting to the immediate right of H.

21. In the above sitting arrangement, which of the following statements is superfluous ?

(*a*) (*i*) (*b*) (*ii*) (*c*) (*iii*) (*d*) (*iv*) (*e*) None is superfluous

22. If E and D, C and B, A and H and K and F interchange their positions, which of the following pairs of students is sitting at the ends ?

(*a*) D and E (*b*) E and F (*c*) D and K

(*d*) K and F (*e*) None of these

Directions (*Questions 23 to 27*) : *Read the following information carefully and answer the questions given below it :* **(Bank P.O. 1995)**

(*i*) Eight persons E, F, G, H, I, J, K and L are seated around a square table — two on each side.

(*ii*) There are three lady members and they are not seated next to each other.

(*iii*) J is between L and F.

(*iv*) G is between I and F.

(*v*) H, a lady member, is second to the left of J.

(*vi*) F, a male member is seated opposite E, a lady member.

(*vii*) There is a lady member between F and I.

23. Who among the following is seated between E and H ?

(*a*) F (*b*) I (*c*) J

(*d*) Cannot be determined (*e*) None of these

24. How many persons are seated between K and F ?

(*a*) One (*b*) Two (*c*) Three

(*d*) Cannot be determined (*e*) None of these

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25. Who among the following are the three lady members ?
(*a*) E, G and J (*b*) E, H and G (*c*) G, H and J
(*d*) Cannot be determined (*e*) None of these

26. Who among the following is to the immediate left of F ?
(*a*) G (*b*) I (*c*) J
(*d*) Cannot be determined (*e*) None of these

27. Which of the following is true about J ?
(*a*) J is a male member. (*b*) J is a female member.
(*c*) Sex of J cannot be determined. (*d*) Position of J cannot be determined.
(*e*) None of these

Directions (*Questions 28-29*) : *On the basis of the information given below, answer questions 28-29.*

(A) P, Q, R, S and T are sitting in a circle facing the centre.
(B) R is immediate left of T.
(C) P is between S and T.

28. Who is to the immediate left of R ?
(*a*) P (*b*) Q (*c*) S (*d*) T (*e*) Cannot be determined

29. To find the answer to the above question, which of the following statements can be dispensed with ?
(*a*) None (*b*) B only (*c*) C only (*d*) B or C only (*e*) None of these

30. Six friends A, B, C, D, E and F are sitting in a closed circle facing the centre. A is facing D. C is between A and B. F is between E and A. Who is to the immediate left of B ? **(I. Tax & Central Excise, 1996)**
(*a*) A (*b*) C (*c*) D (*d*) E

Directions (*Questions 31-32*) : *Read the following information carefully and answer the questions that follow :* **(Bank P.O. 1995)**

A, B, C, D, E and F are seated in a circle facing the centre. D is between F and B. A is second to the left of D and second to the right of E.

31. Who is facing A ?
(*a*) B (*b*) D (*c*) F
(*d*) Either F or B (*e*) None of these

32. Who among the following is facing D ?
(*a*) A (*b*) C (*c*) E
(*d*) Cannot be determined (*e*) None of these

Directions : *On the basis of the information given below, answer questions 33-34.* **(S.B.I.P.O. 1995)**

Eight friends A, B, C, D, E, F, G and H are sitting in a circle facing the centre. B is sitting between G and D. H is third to the left of B and second to the right of A. C is sitting between A and G and B and E are not sitting opposite to each other.

33. Who is third to the left of D ?
(*a*) A (*b*) E (*c*) F
(*d*) Cannot be determined (*e*) None of these

34. Which of the following statements is not correct ?
(*a*) C is third to the right of D. (*b*) A is sitting between C and F.

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(*c*) D and A are sitting opposite of each other.
(*d*) E is sitting between F and D.
(*e*) E and C are sitting opposite of each other.

35. A group of eight members sit in a circle. D is between A and F and is opposite to G. E is to the right of A but on the left of C, whose right hand neighbour is G. B enjoys having H to his left and F to his right. Find the member who is diagonally opposite to A. **(Central Excise, 1995)**

(*a*) B (*b*) F (*c*) G (*d*) H

Directions (*Questions 36 to 38*) : *Study the given information carefully and answer the questions that follow :* **(U.T.I. 1993)**

(*i*) There are seven books one each on Psychology, Hindi, English, Sociology, Economics, Education and Accountancy, lying on a table one above the other.
(*ii*) Sociology is on the top of all the books.
(*iii*) Accountancy is immediately below Education which is immediately below Sociology.
(*iv*) Economics is immediately above Psychology but not in the middle.
(*v*) Hindi is immediately below Psychology.

36. Economics is between which of the following books ?

(*a*) Accountancy and Education (*b*) Psychology and Hindi
(*c*) English and Psychology (*d*) Psychology and Sociology
(*e*) None of these

37. Which three books are between Accountancy and Hindi ?

(*a*) English, Economics and Psychology
(*b*) Economics, Psychology and Education
(*c*) Economics, Psychology and Hindi
(*d*) Cannot be determined
(*e*) None of these

38. If Sociology and English, Accountancy and Hindi and Education and Psychology interchange their positions, which book will be between Psychology and Sociology ?

(*a*) Accountancy (*b*) Psychology (*c*) Hindi
(*d*) Economics (*e*) None of these

39. In a shop, the items were arranged in a shelf consisting of six rows. Biscuits are arranged above the tins of chocolates but below the rows of packets of chips, cakes are at the bottom and the bottles of peppermints are below the chocolates. The topmost row had the display of jam bottles. Where exactly are the bottles of peppermints ? Mention the place from the top. **(Central Excise, 1996)**

(*a*) 2nd (*b*) 3rd (*c*) 4th (*d*) 5th

40. In a pile of reading material, there are novels, story-books, dramas and comics. Every novel has a drama next to it, every story-book has a comic next to it and there is no story-book next to a novel. If there be a novel at the top and the number of books be 40, the order of the books in the pile is :

(*a*) nscd (*b*) ndsc (*c*) csdn (*d*) dncs

(Hotel Management, 1996)

Directions (*Questions 41-42*) : *The following questions are based on the information given below :* **(S.B.I.P.O. 1995)**

(*i*) Seven books are placed one above the other in a particular way.

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(*ii*) History book is placed exactly above Civics book.

(*iii*) Geography book is fourth from the bottom and English book is fifth from the top.

(*iv*) There are two books in between Civics and Economics books.

41. How many books are there between Civics and Science books ? To answer this question, which other extra information is required, if any, from the following ?

(*a*) There are two books between Geography and Science books.

(*b*) There are two books between Mathematics and Geography books.

(*c*) There is one book between English and Science books.

(*d*) The Civics book is before two books above Economics book.

(*e*) No other information is required.

42. Out of the following, which three books are kept above English book ? To answer this question, which of the other informations, if any, is required ?

(*a*) The Economics book is between English and Science books.

(*b*) There are two books between English and History books.

(*c*) The Geography book is above English book.

(*d*) The Science book is placed at the top.

(*e*) No other information is required.

43. In a pile of 10 books, there are 3 of History, 3 of Hindi, 2 of Mathematics and 2 of English. Taking from above, there is an English book between a History and Mathematics book, a History book between a Mathematics and an English book, a Hindi book between an English and a Mathematics book, a Mathematics book between two Hindi books and two Hindi books between a Mathematics and a History book. Book of which subject is at the sixth position from the top ?

(*a*) English (*b*) Hindi (*c*) Mathematics (*d*) History

Directions (*Questions 44 to 48*) : *Read the following information carefully and answer the questions given below it :* **(Bank P.O. 1991)**

In a car exhibition, seven cars of seven different companies *viz.* Cardilac, Ambassador, Fiat, Maruti, Mercedes, Bedford and Fargo were displayed in a row, facing east such that :

(1) Cardilac car was to the immediate right of Fargo.

(2) Fargo was fourth to the right of Fiat.

(3) Maruti car was between Ambassador and Bedford.

(4) Fiat, which was third to the left of Ambassador car, was at one of the ends.

44. Which of the following was the correct position of the Mercedes ?

(*a*) Immediate right of Cardilac (*b*) Immediate left of Bedford

(*c*) Between Bedford and Fargo (*d*) Fourth to the right of Maruti

(*e*) None of these

45. Which of the following is definitely true ?

(*a*) Fargo car is between Ambassador and Fiat.

(*b*) Cardilac car is to the immediate left of Mercedes.

(*c*) Fargo is to the immediate right of Cardilac.

(*d*) Maruti is fourth to the right of Mercedes.

(*e*) None of these

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46. Which cars are on the immediate either sides of the Cardilac car ?
(*a*) Ambassador and Marufi (*b*) Maruti and Fiat (*c*) Fiat and Mercedes
(*d*) Ambassador and Fargo (*e*) None of these

47. Which of the following is definitely true ?
(*a*) Maruti is to the immediate left of Ambassador.
(*b*) Bedford is to the immediate left of Fiat.
(*c*) Bedford is at one of the ends.
(*d*) Fiat is second to the right of Maruti.
(*e*) None of these

48. Which of the following groups of cars is to the right of the Ambassador car ?
(*a*) Cardilac, Fargo and Maruti (*b*) Maruti, Bedford and Fiat
(*c*) Mercedes, Cardilac and Fargo (*d*) Bedford, Cardilac and Fargo

Directions (*Questions 49 to 53*) : *Study the following information carefully and answer the questions given below it :*

All the roads of a city are either perpendicular or parallel to one another. The roads are all straight. Roads A, B, C, D and E are parallel to one another. Roads G, H, I, J, K, L and M are parallel to one another.

(*i*) Road A is 1 km east of road B.
(*ii*) Road B is $\frac{1}{2}$ km west of road C.
(*iii*) Road D is 1 km west of road E.
(*iv*) Road G is $\frac{1}{2}$ km south of road H.
(*v*) Road I is 1 km north of road J.
(*vi*) Road K is $\frac{1}{2}$ km north of road L.
(*vii*) Road K is 1 km south of road M.

49. Which is necessarily true ?
(*a*) E and B intersect. (*b*) D is 2 km west of B.
(*c*) D is at least 2 km west of A. (*d*) M is 1.5 km north of L.
(*e*) I is 1 km north of L.

50. If E is between B and C, which of the following is false ?
(*a*) D is 2 km west of A.
(*b*) C is less than 1.5 km from D.
(*c*) Distance from E to B added to distance of E to C is $\frac{1}{2}$ km.
(*d*) E is less than 1 km from A.
(*e*) D is less than 1 km from B.

51. If road E is between B and C, then distance between A and D is :
(*a*) $\frac{1}{2}$ km (*b*) 1 km (*c*) 1.5 km (*d*) 1.5-2 km (*e*) 2-2.5 km

52. Which of the following possibilities would make two roads coincide ?
(*a*) L is $\frac{1}{2}$ km north of I. (*b*) C is 1 km west of D.
(*c*) I is $\frac{1}{2}$ km north of K. (*d*) D is $\frac{1}{2}$ km east of A.
(*e*) E and B are $\frac{1}{2}$ km apart.

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53. If K is parallel to I and K is $\frac{1}{2}$ km south of J and 1 km north of G, which two roads would be $\frac{1}{2}$ km apart ?

(*a*) I and K (*b*) J and G (*c*) I and G (*d*) J and H (*e*) K and J

Directions (*Questions 54 to 58*) : *Read the following information carefully and answer the questions given below it :* **(M.A.T. 1997)**

Seven friends Kamla, Manish, Rohit, Amit, Gaurav, Pritam and Priya are sitting in a circle. Kamla, Manish, Rohit, Amit, Pritam and Priya are sitting at equal distances from each other.

Rohit is sitting two places right of Pritam, who is sitting one place right of Amit. Kamla forms an angle of 90 degrees from Gaurav and an angle of 120 degrees from Manish. Manish is just opposite Priya and is sitting on the left of Gaurav.

54. Who is the only person sitting between Rohit and Manish ?

(*a*) Pritam (*b*) Amit (*c*) Gaurav (*d*) Kamla

55. Gaurav is not sitting at equal distances from

(*a*) Rohit and Pritam (*b*) Amit and Kamla

(*c*) Manish and Pritam (*d*) All of the above

56. Gaurav is sitting ..... of Priya.

(*a*) to the left (*b*) to the right (*c*) two places right (*d*) None of these

57. The angle between Gaurav and Manish in the clockwise direction is

(*a*) 150° (*b*) 180° (*c*) 210° (*d*) None of these

58. Which of the following statements is not correct ?

(*a*) Pritam is between Manish and Kamla.

(*b*) Manish is two places away from Priya.

(*c*) Gaurav is sitting opposite Pritam.

(*d*) All of the above

ANSWERS

1. (*d*) : Shikha is to the left of Reena and Manju is to her right. Rita is between Reena and Manju. So, the order is : Shikha, Reena, Rita, Manju. In the photograph, Rita will be second from left.

2. (*a*) : B is to the right of D. A is to the right of B. E is to the right of A and left of C. So, the order is : D, B, A, E, C.

Clearly, A is in the middle.

3. (*b*) : Q is left to R and to the right of P *i.e.* P, Q, R.

O is to the right of N and left of P *i.e.* N, O, P.

S is to the right of R and left of T *i.e.* R, S, T.

So, the order is : N, O, P, Q, R, S, T.

Clearly, Q is in the middle.

4. (*d*) : S is sitting next to P. So, the order S, P or P, S is followed. K is sitting next to R. So, the order R, K is followed because R is on the extreme left. T is not next to P or K.

So, the arrangement will be R, K, P, S, T.

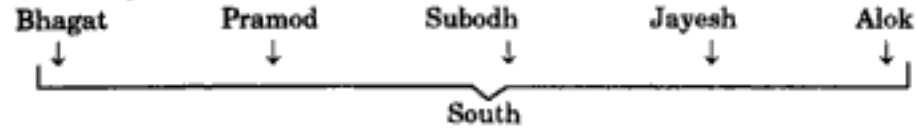
Clearly, P and T are sitting adjacent to S.

5. (*a*) : Clearly, the order is : Anuradha, Rashi, Monika, Sulekha, Abha. Anuradha is adjacent to Rashi.

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Questions 6 to 8

Note : The boys are standing facing south. So, consider 'left' and 'right' accordingly.
Jayesh is to the right of Alok *i.e.* J, A.
Pramod is between Bhagat and Subodh *i.e.* B, P, S.
Subodh is between Jayesh and Pramod.
So, the sequence is :



6. (*a*) : Alok is at the extreme left end.
7. (*d*) : Subodh is in the middle.
8. (*b*) : Statement (A) is superfluous.
9. (*d*) : D is on the left of C *i.e.* C, D.
B is on the right of E *i.e.* B, E.
A is on the right of C *i.e.* A, C.
B is on the left of D *i.e.* D, B.
From the above statements, the correct order is : A, C, D, B, E.
Clearly, D is sitting in the centre.
Note : It is given that A, B, C, D, E are sitting facing you. So, your right and left will be considered as left and right respectively.

Questions 10 to 14

C is to the right of D.
D is third from south. So, B will be at the extreme end from north because it should have E as its neighbour. G is between E and F. So, the sequence is :

B →
E →
G →
F → East
D →
C →
A →

10. (*e*) : G is sitting to the right of E.
11. (*a*) : A and B are sitting at the extreme ends.
12. (*c*) : G should change place with C to make it third from north.
13. (*d*) : D is sitting between C and F.
14. (*d*) : All the statements are required to determine the correct sequence.
15. (*a*) : Clearly, the correct sequence is :
France, America, India, Australia, Japan, China.
The two flags in the centre are of India and Australia.
16. (*b*) : Clearly, C and D sit opposite to each other. So, if A sits opposite to cook, B shall be the cook.
Now, B is to the right of barber. So, one of the rest, say C will be barber. Clearly, then D on the opposite side shall be washerman or tailor. But, washerman is left of tailor and D is to the left of A. So, D is washerman and A is tailor. Thus, A and B are Tailor and Cook.
17. (*c*) : Mr. X is on the right of Mr. Y and on the left of person in the centre *i.e.* Y, X, —
Mr. P is on the right of Mr. Z and Mr. R is on the right of Mr. P *i.e.* Z, P, R.
R is second from the person sitting in the centre.
So, the order is Y, X, Z, P, R. Clearly, Z is sitting in the centre.

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Questions 18 to 22

D is to the left of F and second to the right of C *i.e.* C — D F.

A is second to the right of E *i.e.* E — A.

J is the immediate neighbour of A and B and third to the left of G *i.e.* A J B — G.

H is to the left of D and third to the right of I *i.e.* I — H D.

The above four orders may be combined to obtain the correct order as under :

E, K, A, J, B, I, G, C, H, D, F

18. (*d*) : I is in the middle.

19. (*b*) : C, H, D, F are to the right of G.

20. (*c*) : B is sitting between J and I.

21. (*e*) : All the statements are needed to determine the correct sequence.

22. (*c*) : The new sequence formed on interchanging seats is :

D, F, H, J, C, I, G, B, A, E, K

D and K are sitting at the ends in the new arrangement.

Questions 23 to 27

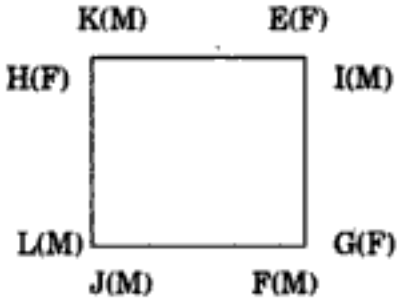
J is between L and F *i.e.* the order is L J F.

G is between I and F *i.e.* the order is F G I.

Thus, the sequence becomes L J F G I.

H is second to the left of J *i.e.* H L J F G I.

Writing the above sequence in form of a square table, we have :



M = Male,
F = Female

E is seated opposite F. Since G is between F and I, so G is a female member.

23. (*e*) : K is seated between E and H.

24. (*c*) : Three persons — H, L and J are seated between K and F.

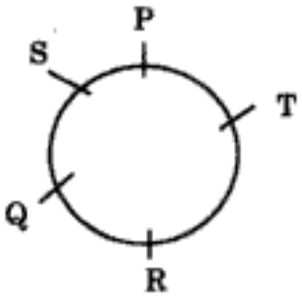
25. (*b*) : The three lady members are E, H and G.

26. (*c*) : J is to the immediate left of F.

27. (*a*) : Clearly, J is a male member.

Questions 28-29

Clearly, in the circle the arrangement is as shown :

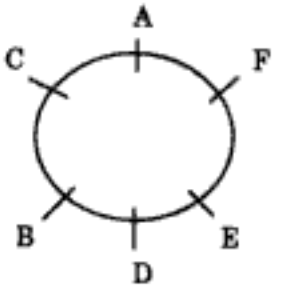


28. (*b*) : Q is to the immediate left of R.

29. (*a*) : All the statements are necessary.

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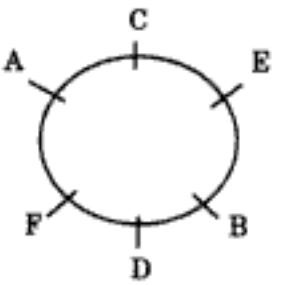
30. (*b*) : Clearly, in a circle the arrangement is as shown.



Thus, C is to the immediate left of B.

Questions 31-32

Clearly, the circular arrangement is as shown :



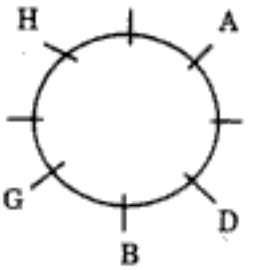
31. (*a*) : Clearly, B is facing A.

32. (*b*) : C is facing D.

Questions 33-34

B is between G and D *i.e.* the order is G B D. H is third to the left of B and second to the right of A.

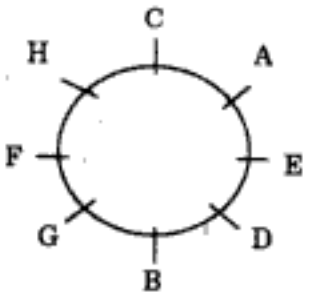
So, forming a circle we have :



C is between A and G. But E is not opposite B.

So, C is between A and H.

Thus, the final arrangement becomes :

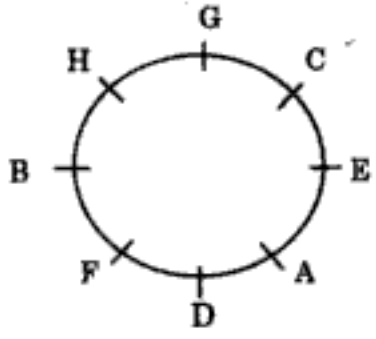


33. (*c*) : F is third to the left of D.

34. (*d*) : Clearly, E is not sitting between F and D.

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35. (*d*) : Clearly, in a circle the arrangement is as shown :



H is diagonally opposite to A.

Questions 36 to 38

Sociology is on top of all books. Education is below Sociology. Accountancy is below Education.

Economics is above Psychology and Psychology is above Hindi.

Economics is not in the middle. So, in the middle lies the seventh book *i.e.* English.

Thus, the sequence from top to bottom is :

Sociology, Education, Accountancy, English, Economics, Psychology, Hindi.

36. (*c*) : Economics is between English and Psychology.

37. (*a*) : There are three books between Accountancy and Hindi — English, Economics, Psychology.

38. (*c*) : The new sequence formed on interchanging places is English, Psychology, Hindi, Sociology, Economics, Education, Accountancy.

Clearly, Hindi is between Psychology and Sociology.

39. (*d*) : Jam bottles are at the top.

Biscuits are below chips, chocolates are below biscuits, peppermints are below chocolates and cakes are at the bottom.

So, the sequence from top to bottom is :

Jam bottles, Chips, Biscuits, Chocolates, Peppermint, Cakes.

40. (*b*) : Clearly, the sequence is : novel, drama, story-book, comic *i.e.* ndsc.

Questions 41-42

Clearly, the sequence of the five books mentioned, from top to bottom is :

——, History, Civics, Geography, English, Economics, ——

41. (*c*) : Clearly, (*c*) gives us the clue that Science book is placed at the bottom. Thus, we know that there are three books between Civics and Science.

42. (*e*) : Clearly, History, Civics and Geography are three books kept above English. To find this, no information other than the given ones is required.

43. (*b*) : Starting from above, English is between History and Mathematics *i.e.*, H, E, M.

History is between Mathematics and English *i.e.*, H, E, M, H, E.

Hindi is between English and Mathematics *i.e.*, H, E, M, H, E, Hi, M.

Mathematics is between two Hindi books *i.e.*, H, E, M, H, E, Hi, M, Hi.

Two Hindi books are between Mathematics and History *i.e.*, H, E, M, H, E, Hi, M, Hi, Hi, H. Clearly, Hindi book is at sixth position from top.

Questions 44 to 48

We analyse the given information as follows :

Arranging the cars from left to right as per the information, we have :

Fargo and Cardilac

Fiat, Car, Car, Car, Fargo

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Maruti between Ambassador and Bedford.

Fiat, Car, Car, Ambassador.

Knowing that Fiat lies at one of the ends, we have from left to right :

1	2	3	4	5	6	7
Fiat	Bedford	Maruti	Ambassador	Fargo	Cardilac	Mercedes

44. (*d*) : Clearly, Maruti is in the third place and Mercedes in the seventh *i.e.* Mercedes is fourth to the right of Maruti.

45. (*b*) : Clearly, the Cardilac on the sixth place, is to the immediate left of the Mercedes, on the seventh place.

46. (*e*) : On either side of the Cardilac are the Fargo and the Mercedes.

47. (*a*) : Clearly, Maruti in the third place, is to the immediate left of Ambassador, in the fourth place.

48. (*c*) : To the right of the Ambassador are the Fargo, Cardilac and Mercedes.

Questions 49 to 53

Clearly from statements (1) and (2), figure 1 follows; from statement (3), figure 2 follows; from statement (4), figure 3 follows; and from statement (5), figure 4 follows; and from statements (6) and (7), figure 5 follows.

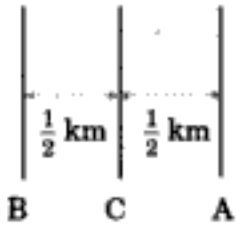


Fig. 1

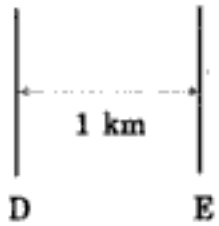


Fig. 2

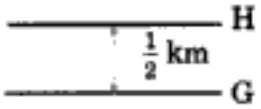


Fig. 3

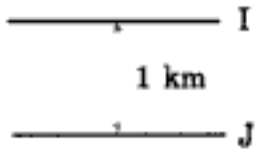


Fig. 4

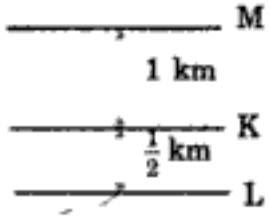
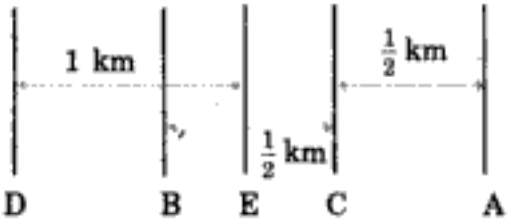


Fig. 5

49. (*d*) : It follows from figure 5 that distance of M from L = LK + KM = $\frac{1}{2}$ + 1 = 1.5 km and M is to the north of L.

50. (*b*) : If E is between B and C, we will have the following figure :



Thus, the statement that D is 2 km west of B is false.

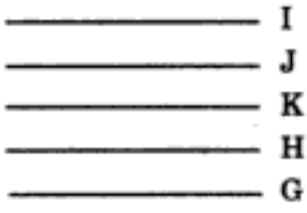
51. (*d*) : From the figure in Q. 50, the distance between A and D

$= AB + ED - BE = \left(1 + 1 - \frac{1}{4}\right) = 2 - .25$ *i.e.* between 1.5 to 2 km.

52. (*e*) : Clearly seeing the figure in Q. 50, we find that if E and B are 1/2 km apart, there is a possibility that E coincides with C.

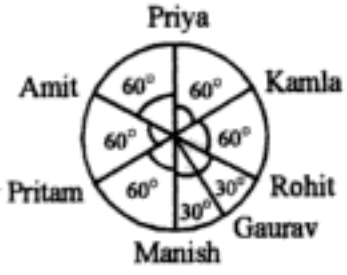
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53. (*e*) : Clearly as per the information combined with figures 3 and 4, the figure is as shown. The two roads J and K, K and H and H and G are 1/2 km apart. So, the answer is K and J.



Questions 54 to 58

Clearly in a circle the arrangement is as shown :



54. (*c*) : Gaurav is sitting between Rohit and Manish.

55. (*d*) : Gaurav is not at equal distances from Rohit and Pritam or Amit and Kamla or Manish and Pritam.

56. (*d*) : Gaurav is three places left or four places right of Priya.

57. (*d*) : The angle between Gaurav and Manish in clockwise direction is 30°.

58. (*d*) : Clearly, all the statements follow from the diagram above.

TYPE 3 : COMPARISON TYPE QUESTIONS

In such type of questions, clues are given regarding comparisons among a set of persons or things with respect to one or more qualities. The candidate is required to analyse the whole information, form a proper ascending/descending sequence and then answer the given questions accordingly.

Ex. 1. ***Read the following information and answer the questions given below it :***

There are five friends — Sachin, Kunal, Mohit, Anuj and Rohan.
Sachin is shorter than Kunal but taller than Rohan.
Mohit is the tallest.
Anuj is a little shorter than Kunal and little taller than Sachin.

1. Who is the shortest ?
(*a*) Rohan (*b*) Sachin (*c*) Anuj (*d*) Kunal (*e*) None of these

2. If they stand in the order of their heights, who will be in the middle ?
(*a*) Kunal (*b*) Rohan (*c*) Sachin (*d*) Anuj (*e*) None of these

3. If they stand in the order of increasing heights, who will be the second ?
(*a*) Anuj (*b*) Sachin (*c*) Rohan (*d*) Kunal (*e*) None of these

4. Who is the second tallest ?
(*a*) Sachin (*b*) Kunal (*c*) Anuj (*d*) Rohan (*e*) None of these

5. Who is taller than Anuj but shorter than Mohit ?
(*a*) Kunal (*b*) Rohan (*c*) Sachin
(*d*) Data inadequate (*e*) None of these

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Solution : Let us denote the five boys by the first letter of their names, namely S, K, M, A and R.

Then, $R < S < K < M$ and $S < A < K$.

$\therefore$ $R < S < A < K < M$

1. (*a*) : Rohan is shortest.

2. (*d*) : Anuj is in the middle.

3. (*b*) : In the order of increasing heights *i.e.* shortest to tallest, Sachin is second.

4. (*b*) : Kunal is second tallest.

5. (*a*) : Kunal is taller than Anuj but shorter than Mohit.

Ex. 2. Read the information given below and answer the questions that follow :

(*i*) There is a group of five girls.

(*ii*) Kamini is second in height but younger than Rupa.

(*iii*) Pooja is taller than Monika but younger in age.

(*iv*) Rupa and Monika are of the same age but Rupa is tallest between them.

(*v*) Neelam is taller than Pooja and elder to Rupa. **(Bank P.O. 1996)**

1. If they are arranged in the ascending order of height, who will be in third position ?

(*a*) Monika (*b*) Rupa (*c*) Monika or Rupa

(*d*) Data inadequate (*e*) None of these

2. If they are arranged in the descending order of their ages, who will be in fourth position ?

(*a*) Monika or Rupa (*b*) Kamini (*c*) Monika

(*d*) Data inadequate (*e*) None of these

3. To answer the question "who is the youngest person in the group", which of the given statements is superfluous ?

(*a*) Only (*i*) (*b*) Only (*ii*) (*c*) Only (*v*)

(*d*) Either (*i*) or (*iv*) (*e*) None of these

Solution : We first find the sequence of heights :

By (*iii*), we have : $M < P$.

By (*v*), we have : $P < N$.

Now, Rupa is tallest and Kamini is second in height.

So, the sequence of heights is : $M < P < N < K < R$.

Now, we determine the age sequence :

By (*ii*), we have : $K < R$.

By (*iii*), we have : $P < M$.

By (*iv*), we have : $R = M$.

By (*v*), we have : $R < N$.

So, the sequence of ages is : $N < R = M < K < P$ or $N < R = M < P < K$.

1. Clearly, in the increasing order of height, Neelam is in third position.

Hence, the answer is (*e*).

2. Clearly, in the descending order of ages, Neelam will be in fourth position (because Monika and Rupa both lie at third position).

Hence, the answer is (*e*).

3. Clearly, only statement (*i*) is not necessary.

Hence, the answer is (*a*).

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EXERCISE 6C

1. Compare the knowledge of persons X, Y, Z, A, B and C in relation to each other :
1. X knows more than A.
2. Y knows as much as B.
3. Z knows less than C.
4. A knows more than Y.

The best knowledgeable person amongst all is : **(S.C.R.A. 1996)**

(*a*) X (*b*) Y (*c*) A (*d*) C

2. Five children were administered psychological tests to know their intellectual levels. In the report, psychologists pointed out that the child A is less intelligent than the child B. The child C is less intelligent than the child D. The child B is less intelligent than the child C and child A is more intelligent than the child E. Which child is the most intelligent ? **(Bank P.O. 1996)**

(*a*) A (*b*) B (*c*) D (*d*) E (*e*) None of these

3. Among five boys, Vineet is taller than Manick, but not as tall as Ravi. Jacob is taller than Dilip but shorter than Manick. Who is the tallest in their group ?

(*a*) Ravi (*b*) Manick (*c*) Vineet
(*d*) Cannot be determined (*e*) None of these **(NABARD, 1994)**

4. If (*i*) P is taller than Q; (*ii*) R is shorter than P; (*iii*) S is taller than T but shorter than Q, then who among them is the tallest ? **(B.S.R.B. 1995)**

(*a*) P (*b*) Q (*c*) S (*d*) T (*e*) Can't be determined

5. Five boys participated in a competition. Rohit was ranked lower than Sanjay. Vikas was ranked higher than Dinesh. Kamal's rank was between Rohit and Vikas. Who was ranked highest ?

(*a*) Sanjay (*b*) Vikas (*c*) Dinesh (*d*) Kamal (*e*) None of these

6. In an examination, Raj got more marks than Mukesh but not as many as Priya. Priya got more marks than Gaurav and Kavita. Gaurav got less marks than Mukesh but his marks are not the lowest in the group. Who is second in the descending order of marks ? **(Bank P.O. 1997)**

(*a*) Priya (*b*) Kavita (*c*) Raj
(*d*) Cannot be determined (*e*) None of these

7. Ashish is heavier than Govind. Mohit is lighter than Jack. Pawan is heavier than Jack but lighter than Govind. Who among them is the heaviest ? **(Bank P.O. 1995)**

(*a*) Govind (*b*) Jack (*c*) Pawan (*d*) Ashish (*e*) Mohit

8. Pune is bigger than Jhansi, Sitapur is bigger than Chittor. Raigarh is not as big as Jhansi, but is bigger than Sitapur. Which is the smallest ? **(Railways, 1994)**

(*a*) Pune (*b*) Jhansi (*c*) Sitapur (*d*) Chittor

9. Rohan is taller than Anand but shorter than Seema. Krishna is taller than Pushpa but shorter than Anand. Dhiraj is taller than Krishna but shorter than Seema. Who among them is the tallest ? **(B.S.R.B. 1996)**

(*a*) Rohan (*b*) Seema (*c*) Krishna
(*d*) Cannot be determined (*e*) None of these

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Directions (*Questions 10-11*) : *Read the following information carefully and answer the questions given below it :* **(Bank P.O. 1994)**

(A) Gopal is shorter than Ashok but taller than Kunal;
(B) Navin is shorter than Kunal;
(C) Jayesh is taller than Navin;
(D) Ashok is taller than Jayesh.

10. Who among them is the tallest ?
(*a*) Gopal (*b*) Ashok (*c*) Kunal (*d*) Navin (*e*) Jayesh

11. Which of the given informations is not necessary to answer the above question ?
(*a*) A (*b*) B (*c*) C (*d*) D (*e*) None of these

12. B is twice as old as A but twice younger than F.
C is half the age of A but twice the age of D.
Which two persons form the pair of oldest and youngest ?
(*a*) F and A (*b*) F and D (*c*) B and F
(*d*) F and C (*e*) None of these

13. Sudhanshu is as much older than Kokila as he is younger than Praveen. Nitin is as old as Kokila. Which of the following statements is wrong ? **(Bank P.O. 1995)**
(*a*) Kokila is younger than Praveen. (*b*) Nitin is younger than Praveen.
(*c*) Sudhanshu is older than Nitin. (*d*) Praveen is not the oldest.
(*e*) Kokila is younger than Sudhanshu.

14. A is elder to B while C and D are elder to E who lies between A and C. If C be elder to B, which one of the following statements is necessarily true ?
(*a*) A is elder to C (*b*) C is elder to D
(*c*) D is elder to C (*d*) E is elder to B
(Hotel Management, 1995)

15. Hitesh is richer than Jaya whereas Mohan is richer than Pritam. Lalit is as rich as Jaya. Amit is richer than Hitesh. What conclusion can be definitely drawn from the above statements ? **(Hotel Management, 1996; Asstt. Grade, 1995)**
(*a*) Jaya is poorer than Pritam. (*b*) Mohan is richer than Amit.
(*c*) Lalit is poorer than Hitesh. (*d*) Pritam is richer than Lalit.

Directions (*Questions 16 to 21*) : *Study the following information and answer the questions given below it :*

A blacksmith has five iron articles A, B, C, D and E, each having a different weight.

(*i*) A weighs twice as much as B.
(*ii*) B weighs four and a half times as much as C.
(*iii*) C weighs half as much as D.
(*iv*) D weighs half as much as E.
(*v*) E weighs less than A but more than C.

16. Which of the following is the lightest in weight ?
(*a*) A (*b*) B (*c*) C (*d*) D (*e*) E

17. E is lighter in weight than which of the other two articles ?
(*a*) A, B (*b*) D, C (*c*) A, C (*d*) D, B (*e*) B, E

18. E is heavier than which of the following two articles ?
(*a*) D, B (*b*) D, C (*c*) A, C (*d*) A, B (*e*) None of these

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19. Which of the following articles is the heaviest in weight ?
(*a*) A (*b*) B (*c*) C (*d*) D (*e*) E

20. Which of the following represents the descending order of weights of the articles ?
(*a*) A, B E, D, C (*b*) B, D, E, A, C (*c*) E, C, D, A, B
(*d*) C, A, D, B, E (*e*) A, B, D, E, C

21. Which of the above given statements is not necessary to determine the correct order of articles according to their weights ?
(*a*) 1 (*b*) 2 (*c*) 3 (*d*) 4 (*e*) 5

Directions (*Questions 22 to 26*) : *Read the following information and answer the questions given below it :*

(*i*) Seven students P, Q, R, S, T, U and V take a series of tests.
(*ii*) No two students get similar marks.
(*iii*) V always scores more than P.
(*iv*) P always scores more than Q.
(*v*) Each time either R scores the highest and T gets the least, or alternatively S scores the highest and U or Q scores the least.

22. If S is ranked sixth and Q is ranked fifth, which of the following can be true ?
(*a*) V is ranked first or fourth. (*b*) R is ranked second or third.
(*c*) P is ranked second or fifth. (*d*) U is ranked third or fourth.
(*e*) T is ranked fourth or fifth.

23. If R gets most, V should be ranked not lower than :
(*a*) second (*b*) third (*c*) fourth (*d*) fifth (*e*) sixth

24. If R is ranked second and Q is ranked fifth, which of the following must be true ?
(*a*) S is ranked third. (*b*) T is ranked sixth. (*c*) P is ranked sixth.
(*d*) V is ranked fourth. (*e*) U is ranked sixth.

25. If S is ranked second, which of the following can be true ?
(*a*) U gets more than V. (*b*) V gets more than S. (*c*) P gets more than R.
(*d*) P gets more than V. (*e*) T gets more than Q.

26. If V is ranked fifth, which of the following must be true ?
(*a*) S scores the highest. (*b*) R is ranked second. (*c*) T is ranked third.
(*d*) Q is ranked fourth. (*e*) U scores the least.

Directions (*Questions 27 to 31*) : *Study the information given below and answer the questions that follow :* **(Bank P.O. 1995)**

(*i*) A, B, C, D, E and F are six students in a class.
(*ii*) B and C are shorter than F but heavier than A.
(*iii*) D is heavier than B and taller than C.
(*iv*) E is shorter than D but taller than F.
(*v*) F is heavier than D.
(*vi*) A is shorter than E but taller than F.

27. Who among them is the tallest ?
(*a*) A (*b*) B (*c*) D (*d*) E (*e*) None of these

28. Who is third from the top when they are arranged in descending order of height ?
(*a*) A (*b*) B (*c*) C (*d*) E (*e*) None of these

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29. Which of the following groups of friends is shorter than A ?
(*a*) B, C only (*b*) D, B, C only (*c*) E, B, C only
(*d*) F, B, C, only (*e*) None of these

30. Who among them is the lightest ?
(*a*) A (*b*) B (*c*) C (*d*) B or C (*e*) Data inadequate

31. Which of the following statements is true for F as regards height and weight ?
(*a*) He is lighter than E and taller than E.
(*b*) He is heavier than B and taller than E.
(*c*) He is heavier than B and C but shorter than D.
(*d*) He is lighter than E and also shorter than E.
(*e*) He is lighter than B and C but taller than D.

Directions (*Questions 32 to 36*) : *Read the following information carefully and answer the questions given below it :*

(*i*) A, B, C, D and E are five friends.
(*ii*) B is elder to E, but not as tall as C.
(*iii*) C is younger to A, and is taller to D and E.
(*iv*) A is taller to D, but younger to E.
(*v*) D is elder to A but is shortest in the group.

32. Who among the following is the eldest ?
(*a*) A (*b*) B (*c*) C (*d*) D (*e*) None of these

33. Which of the following pairs of students is elder to D ?
(*a*) BA (*b*) BC (*c*) BE (*d*) EA (*e*) None of these

34. Which of the following statements is correct about B ?
(*i*) B is not the tallest.
(*ii*) B is shorter to E.
(*iii*) When they are asked to stand in ascending order with respect to their heights, B is in the middle.
(*a*) Only (*i*) is correct (*b*) Only (*i*) and (*ii*) are correct (*c*) All are correct
(*d*) All are incorrect (*e*) None of these

35. If F, another friend, is taller than C, how many of them will be between F and E according to their height ?
(*a*) None (*b*) One (*c*) Two (*d*) Three (*e*) None of these

36. If a selection is to be made among them who would be relatively older and also taller, who among them should be chosen ?
(*a*) A (*b*) B (*c*) C (*d*) D (*e*) E

Directions (*Questions 37 to 41*) : *Study the given information carefully and answer the questions that follow :* **(Bank P.O. 1994)**

(*i*) Six friends P, Q, R, S, T and U are members of a club and play a different game of Football, Cricket, Tennis, Basketball, Badminton and Volleyball.
(*ii*) T who is taller than P and S plays Tennis.
(*iii*) The tallest among them plays Basketball.
(*iv*) The shortest among them plays Volleyball.
(*v*) Q and S neither play Volleyball nor Basketball.
(*vi*) R plays Volleyball.
(*vii*) T is between Q who plays Football and P in order of height.

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37. Who among them is taller than R but shorter than P ?
(*a*) Q (*b*) T (*c*) U (*d*) Data inadequate (*e*) None of these

38. Who will be at the third place if they are arranged in the descending order of their height ?
(*a*) Q (*b*) P (*c*) S (*d*) T (*e*) None of these

39. Which of the following statements is not true ?
(*a*) P is shorter than R. (*b*) Q is taller than S. (*c*) S is taller than R.
(*d*) T is taller than R. (*e*) U is taller than Q.

40. Who among them plays Basketball ?
(*a*) Q (*b*) R (*c*) S (*d*) U (*e*) None of these

41. What does S play ?
(*a*) Cricket (*b*) Badminton (*c*) Football
(*d*) Either Cricket or Badminton (*e*) None of these

Directions* (*Questions 42-43*) : *Read the following information carefully and answer the questions given below it : **(Bank P.O. 1997)**

At the end of a cricket series, when five players were arranged in the ascending order of runs scored by them, O was fourth while N was first. When they were arranged in descending order for wickets taken by them, K replaces O while O replaces L. M's position remains unchanged. K has scored more runs than M. L is having first rank in one ranking and fifth in another.

42. Who has scored the highest runs in the series ?
(*a*) K (*b*) L (*c*) M (*d*) Can't be determined (*e*) None of these

43. Who has taken the lowest number of wickets ?
(*a*) L (*b*) M (*c*) P (*d*) Can't be determined (*e*) None of these

Directions (*Questions 44-47*) : *Read the information given below and answer the questions that follow :* **(M.B.A. 1997)**

A * B means A and B are of the same age;
A – B means B is younger than A;
A + B means A is younger than B.

44. Sachin * Madan – Reena means
(*a*) Reena is the youngest. (*b*) Reena is the oldest.
(*c*) Madan is younger than Reena. (*d*) None of these

45. X + Y + Z is same as
(*a*) Y – X – Z (*b*) Z – Y – X (*c*) Z – X – Y (*d*) None of these

46. For an expression Farha – Farida – Arif, which of the following cannot be correct under any circumstances ?
(*a*) Arif is the father of Farha.
(*b*) Arif is the younger brother of Farha.
(*c*) Farha is the mother of both Arif and Farida.
(*d*) None of these

47. Deven – Shashi * Hemant is opposite to
(*i*) Hemant + Shashi + Deven (*ii*) Hemant – Shashi + Deven
(*iii*) Shashi * Hemant + Deven
(*a*) (*i*) only (*b*) (*i*) and (*ii*) only (*c*) (*ii*) and (*iii*) only (*d*) None of these

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ANSWERS

1. (*a*) : Clearly, we have : $A < X$, $Y = B$, $Z < C$, $Z < B$, $Y < A$.
Thus, the sequence becomes :
$X > A > Y = B > C > Z$.
So, X is the best knowledgeable person.

2. (*c*) : We have : $A < B$, $C < D$, $B < C$ and $E < A$.
So, the sequence becomes : $E < A < B < C < D$.
Clearly, child D is the most intelligent.

3. (*a*) : In terms of height, we have :
Manick < Vineet, Vineet < Ravi, Dilip < Jacob, Jacob < Manick.
So, the sequence becomes :
Dilip < Jacob < Manick < Vineet < Ravi.
Clearly, Ravi is the tallest.

4. (*a*) : In terms of height, we have :
$Q < P$, $R < P$, $T < S$, $S < Q$.
So, the sequence becomes : $T < S < Q < R < P$ or $T < S < R < Q < P$.
Whichever may be the case, P is the tallest.

5. (*a*) : In terms of rank, we have :
Rohit < Sanjay, Dinesh < Vikas.
Since Kamal's rank is between Rohit and Vikas, the sequence becomes :
Dinesh < Vikas < Kamal < Rohit < Sanjay.
Clearly, Sanjay was ranked highest.

6. (*c*) : In terms of marks obtained,
Mukesh < Raj, Raj < Priya, Gaurav < Priya, Kavita < Priya, Gaurav < Mukesh.
Since Gaurav's marks are not the lowest, so Kavita's marks are the lowest.
So, the sequence becomes :
Kavita < Gaurav < Mukesh < Raj < Priya.
Clearly, in the descending order, Raj comes second.

7. (*d*) : In terms of weight, we have :
Govind < Ashish, Mohit < Jack, Jack < Pawan, Pawan < Govind.
So, the sequence becomes :
Mohit < Jack < Pawan < Govind < Ashish.
Clearly, Ashish is the heaviest.

8. (*d*) : In terms of size, we have :
Jhansi < Pune, Chittor < Sitapur, Raigarh < Jhansi, Sitapur < Raigarh.
So, the sequence becomes :
Chittor < Sitapur < Raigarh < Jhansi < Pune.
Clearly, Chittor is the smallest.

9. (*b*) : In terms of height, we have :
Anand < Rohan, Rohan < Seema, Pushpa < Krishna, Krishna < Anand.
Krishna < Dhiraj, Dhiraj < Seema.
So, the sequence becomes :
Pushpa < Krishna < Dhiraj < Anand < Rohan < Seema.
Clearly, Seema is the tallest.

10. (*b*) : In terms of height, we have :
Gopal < Ashok, Kunal < Gopal, Navin < Kunal, Navin < Jayesh, Jayesh < Ashok.

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So, the sequence becomes :
Navin < Kunal < Gopal < Jayesh < Ashok.
Clearly, Ashok is the tallest.

11. (*c*) : Clearly, statement C is not necessary.

12. (*b*) : Let A's age be x. Then, B's age is $2x$. B is twice younger than F *i.e.* F is twice older than B. So, F's age is $4x$. C is half the age of A *i.e.* C's age is $\frac{x}{2}$. C is twice the age of D *i.e.* D is half the age of C *i.e.* D's age is $\frac{x}{4}$. So, the descending order of ages is F, B, A, C, D.
Clearly, F is the oldest and D is the youngest.

13. (*d*) : In terms of age, we have :
Kokila < Sudhanshu, Sudhanshu < Praveen, Nitin = Kokila.
So, the sequence becomes :
Nitin = Kokila < Sudhanshu < Praveen.
Clearly, Praveen is the oldest.
Hence, (*d*) is the incorrect statement.

14. (*d*) : In terms of age, we have :
B < A, E < C, E < D, B < C.
Since E lies between A and C, the sequence becomes :
B < A < E < C < D or B < A < E < D < C
Clearly, whichever may be the case, E is elder to B. Hence, (*d*) is necessarily true.

15. (*c*) : In terms of richness, we have :
Jaya < Hitesh, Pritam < Mohan, Lalit = Jaya, Hitesh < Amit.
So, we have : Lalit = Jaya < Hitesh < Amit and Pritam < Mohan.
Clearly, Lalit is poorer than Hitesh.

Questions 16 to 21

Let C's weight be x. Then, D's weight is $2x$. E's weight is $4x$, B's weight is $4.5x$ and A's weight is $9x$.
So, the order of weights can be A > B > E > D > C.

16. (*c*) : C is the lightest in weight.

17. (*a*) : E is lighter in weight than A and B.

18. (*b*) : E is heavier than D and C.

19. (*a*) : Clearly, A is the heaviest in weight.

20. (*a*) : Clearly, the descending order of weights is A, B, E, D, C.

21. (*e*) : Clearly, statement (5) is not required to determine the order of weights.

Questions 22 to 26

In terms of scores, we have :
V > P, P > Q *i.e.* V > P > Q.
If R scores the highest, we have R > ..... > T.
If S scores the highest, we have S > ..... > Q or S > ..... > U.

22. (*d*) : If S is ranked sixth and Q is ranked fifth, we have :
□ > □ > □ > □ > Q > S > □ .
In this case, R will rank the highest and thus T will rank the least. We have :
R > □ > □ > □ > Q > S > T.
Also, the order V > P > Q will be maintained *i.e.*, V and P will have second, third or fourth places. So, statements (*a*), (*b*), (*c*) and (*e*) cannot follow. Thus, (*d*) is the answer.

23. (*c*) : Again, if R ranks most, T ranks lowest and occupies seventh place. Since V always ranks above P and Q, so in the maximum, P and Q will occupy fifth and sixth places. Thus, V will not rank lower than fourth.

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24. (*b*) : If R is ranked second, S will rank first and Q and U lowest. But Q ranks fifth. So, U ranks lowest. Also, the order V > P > Q will be followed.

So, the arrangement will be S > R > V > P > Q > □ > U. Thus, the sixth place will be occupied by T.

25. (*a*) : If S ranks second, R ranks first and T ranks lowest. The order V > P > Q will be followed. So, the arrangement will be R > S > □ > □ > □ > □ > T.

Clearly, statements (*b*), (*c*), (*d*) and (*e*) cannot follow. So, the answer is (*a*).

26. (*a*) : If V ranks fifth, P and Q coming before it will occupy sixth and seventh places respectively *i.e.* Q ranks least. So, S will score the highest.

Questions 27 to 31

In terms of height, we have :

B < F, C < F, C < D, E < D, F < E, A < E, F < A.

So, C < F < E < D, B < F, F < A < E

Thus, the sequence becomes :

B < C < F < A < E < D or C < B < F < A < E < D.

In terms of weight, we have :

A < B, A < C, B < D, D < F.

So, A < B < D < F, A < C.

Thus, the sequence becomes :

A < C < B < D < F or A < B < C < D < F or A < B < D < C < F.

27. (*c*) : Clearly, D is the tallest.

28. (*a*) : The descending order of height is :

D > E > A > F > B > C or D > E > A > F > C > B.

Clearly, A is third from the top.

29. (*d*) : Clearly, F, B and C are shorter than A.

30. (*e*) : Data is inadequate as no clue regarding E's weight is given.

31. (*c*) : Clearly F is heavier than B and C but shorter than D.

Questions 32 to 36

In terms of age, we have :

E < B, C < A, A < E, A < D

So, we have : C < A < E < B, A < D.

In terms of height, we have :

B < C, D < C, E < C, D < A.

32. (*e*) : Either B or D is the eldest.

33. (*e*) : It cannot be determined for sure.

34. (*a*) : B is shorter than C. So, B is not the tallest. Thus, (*i*) is correct.

B and E are shorter than C. So, it cannot be concluded that B is shorter to E. Thus, (*ii*) is incorrect.

A single definite order of heights cannot be obtained from the given information. So, (*iii*) is incorrect.

35. (*e*) : Since no definite order of height can be obtained, so it cannot be determined for sure how many persons lie between F and E.

36. (*b*) : A and C are youngest so they cannot be selected. D is shorter than two persons A and C.

B is shorter than C only and is also relatively older. So, B will be selected.

E is younger than B.

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Questions 37 to 41

In terms of height, we have

T > P, T > S, Q > T > P.

R plays Volleyball, so R is the shortest.

Q plays neither Volleyball nor Basketball.

So, Q is not the tallest. Thus, U is the tallest.

So, the sequence becomes : U > Q > T > P > S > R.

Now, T plays Tennis. U, being tallest, plays Basketball. R plays Volleyball. Q plays Football. Both P and S play either Cricket or Badminton.

37. (*e*) : S is taller than R but shorter than P.

38. (*d*) : The descending order of height is U, Q, T, P, S, R.

Clearly, T is at the third place.

39. (*a*) : Clearly, P is taller than R.

40. (*d*) : U plays Basketball.

41. (*d*) : S plays either Cricket or Badminton.

Questions 42-43

In terms of runs scored, we have the ascending order as N < □ < □ < O < □ .

N has the first rank. So, L will be fifth in this order *i.e.*, N < □ < □ < O < L.

K has scored more runs than M *i.e.*, K > M.

So, the sequence becomes N < M < K < O < L.

In terms of wickets taken, the order becomes : L > M > N > K > O.

42. (*b*) : L has scored the highest runs in the series.

43. (*e*) : O has taken the lowest number of wickets.

44. (*a*) : Sachin * Madan – Reena means Sachin and Madan are of the same age and Reena is younger than Madan. This means that Reena is the youngest.

45. (*b*) : X + Y + Z means X is younger than Y and Y is younger than Z. This can also be written as Z – Y – X.

46. (*a*) : Farha – Farida – Arif means Farida is younger than Farha and Arif is younger than Farida. This means that Arif is younger than Farha. So, Arif cannot be the father of Farha.

47. (*d*) : Deven – Shashi * Hemant means Shashi is younger than Deven, and Shashi and Hemant are of the same age. Thus, Deven is the oldest. Now, the opposite statement would mean : Deven is the youngest.

(*i*) Hemant + Shashi + Deven means Hemant is younger than Shashi, who is younger than Deven. So, Deven is the oldest.

(*ii*) Hemant – Shashi + Deven means Shashi is younger than both Hemant and Deven. Thus, either Hemant or Deven is the oldest, but Deven is not the youngest.

(*iii*) Shashi * Hemant + Deven means Shashi and Hemant are of the same age and Hemant is younger than Deven. So, Deven is the oldest.

TYPE 4 : SEQUENTIAL ORDER OF THINGS

In this type of questions, some clues are given regarding the order of occurrence of certain events. The candidate is required to analyse the given information, frame the right sequence and then answer the questions accordingly.

Ex. 1. Read the following information and answer the questions given below it :

Six plays — A, B, C, D, E and F are to be staged, one on each day from Monday to Saturday. The schedule of the plays is to be in accordance with the following :

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(*i*) A must be staged a day before E.
(*ii*) C must not be staged on Tuesday.
(*iii*) B must be staged on the day following the day on which F is staged.
(*iv*) D must be staged on Friday only and should not be immediately preceded by B.
(*v*) E must not be staged on the last day of the schedule.

1. Which of the following plays immediately follows B ?
(*a*) A (*b*) C (*c*) D (*d*) E (*e*) F

2. Which of the following plays is on Monday ?
(*a*) E (*b*) F (*c*) C (*d*) B (*e*) A

3. Play D is between which of the following pairs of plays ?
(*a*) B and E (*b*) E and F (*c*) A and E (*d*) C and E (*e*) C and F

4. Which of the following is the schedule of plays with the order of their staging from Monday ?
(*a*) E, A, B, F, D, C (*b*) A, F, B, E, D, C (*c*) A, F, B, C, D, E
(*d*) F, A, B, E, D, C (*e*) None of these

5. Play C cannot definitely be staged on which of the following days in addition to Tuesday ?
(*a*) Monday (*b*) Wednesday (*c*) Friday (*d*) Thursday (*e*) Saturday

---

Solution : Clearly, D must be staged on Friday. A must be staged before E *i.e.*, order AE must be followed. But E cannot be staged on last day. Also, B must be staged immediately after F *i.e.*, order FB must be followed. But B cannot precede D. So, F and B can be staged on Monday and Tuesday and A and E on Wednesday and Thursday. C, which cannot be staged on Tuesday shall be staged on Saturday. Thus, the order followed will be :

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
F	B	A	E	D	C

1. Clearly, A immediately follows B. So, the answer is (*a*).
2. F will be played on Monday. So, the answer is (*b*).
3. Play D is between E and C. So, the answer is (*d*).
4. Clearly, order of staging of plays is F, B, A, E, D, C. So, the answer is (*e*).
5. C cannot be staged on Friday as well because D has to be staged on that day.

Ex. 2. Read the following information carefully and answer the questions given below it : **(S.B.I.P.O. 1997)**

(*i*) Eight doctors P, Q, R, S, T, U, V and W visit a charitable dispensary every day except on a holiday *i.e.* Monday.
(*ii*) Each doctor visits for one hour from Tuesday to Sunday except Saturday. The timings are 9 a.m. to 1 p.m. and 2 p.m. to 6 p.m.; 1 p.m. to 2 p.m. is lunch break.
(*iii*) On Saturday, it is open only in the morning *i.e.* 9 a.m. to 1 p.m. and each doctor visits for only half an hour.
(*iv*) No other doctor visits the dispensary before doctor Q and after doctor U.
(*v*) Doctor W comes immediately after lunch break and is followed by R.
(*vi*) S comes in the same order as P in the afternoon session.

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